

Experiment 12: Hill Cipher Encryption and Decryption:

```
import numpy as np
```

```
k=np.array([[3,3],[2,5]])  
print("Key Matrix:\n",k)
```

```
p=[ord(x)-65 for x in input("Enter Plaintext: ").upper()]  
c=np.dot(k,p)%26
```

```
print("Ciphertext:", ".join(chr(int(x)+65) for x in c))
```

```
ki=np.array([[15,17],[20,9]])  
d=np.dot(ki,c)%26
```

```
print("Decrypted Text:", ".join(chr(int(x)+65) for x in d))
```

Output:

```
Key Matrix:  
[[3 3]  
[2 5]]  
Enter Plaintext: hi  
Ciphertext: TC  
Decrypted Text: HI  
  
=== Code Execution Successful ===
```